

Robust Phishing Detection via DOM Structural Geometry and Mid-Fusion XGBoost

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Abstract—As phishing attacks grow increasingly sophisticated, traditional lexical and content-based detection models face severe degradation over time due to adversarial evasion and domain shift. Attackers easily manipulate URLs and superficial HTML content to bypass advanced Deep Learning (DL) architectures. In this study, we propose an alternative, highly robust methodology focusing on the immutable structural geometry of the Document Object Model (DOM). We extracted 163 multimodal features (10 URL lexical metrics, 18 DOM geometric metrics, and 135 TF-IDF structural metrics comprising 50 standard sequence features, 60 spatial zone features, and 25 hierarchical XPath traces) utilizing Python’s `BeautifulSoup` to map the inherent architectural differences between legitimate websites and the superficial structural elements of phishing pages. We collected three distinct datasets representing temporal shifts: Dataset-2021, Dataset-2024, and Dataset-2026. We evaluated traditional NLP models, state-of-the-art DL models (including WebPhish CNN), and our proposed Mid-Fusion XGBoost architecture. Experimental results demonstrate that while DL models operating on raw text experience significant degradation on new domains (dropping from 98.9% local accuracy to 55.5% on Dataset-2024), our structural Mid-Fusion model retains a robust 79.32% accuracy on Dataset-2024, and achieves 79.83% accuracy on Dataset-2026 (albeit with reduced precision under heavy sampling bias). Furthermore, we demonstrate that structural Machine Learning approaches train $6\times$ faster than Deep Learning models (5 minutes vs. 30 minutes), making them exceptionally suited for the rapid retraining required in the highly volatile cybersecurity landscape. We employ t-SNE to quantify the domain shift and utilize SHAP to provide explainable AI insights into feature importance drift.

Index Terms—Phishing Detection, Domain Shift, XGBoost, DOM Structure, Explainable AI, SHAP, Zero-Day, Deep Learning

I. INTRODUCTION

Phishing remains one of the most pervasive cybersecurity threats, serving as the primary vector for malware distribution, credential theft, and ransomware attacks. Despite decades of research and the advent of sophisticated Deep Learning classifiers, phishing is a fundamentally moving target. Modern detection models often fail in real-world deployment, not simply due to generic “modern attacks,” but because of severe domain/content shift and active adversarial bypassing. Attackers operate as advanced persistent threats (APTs) and utilize Phishing-as-a-Service (PhaaS) platforms to actively study the parameters of security filters. They deliberately alter URLs

and scrape raw HTML text from legitimate targets to mask their malicious intent.

Consequently, models that over-rely on lexical analysis or purely textual Deep Learning representations inevitably decay. A classical model trained to heavily penalize the absence of HTTPS will suddenly fail when attackers universally adopt free SSL certificates (such as Let’s Encrypt). Similarly, models that embed the raw HTML corpus using Convolutional Neural Networks (CNNs) can be trivially bypassed when the attacker copies the target’s exact HTML code and injects a single malicious iframe.

A. The Necessity of Rapid Retraining

A critical limitation of advanced Deep Learning (DL) approaches in phishing detection is the immense computational overhead required for training. As adversarial tactics evolve on a daily basis, cybersecurity filters must be retrained constantly to patch newly discovered vulnerabilities. DL models often require specialized GPU hardware and hours of training time, leading to unacceptable vulnerability windows. Conversely, tree-based Machine Learning (ML) models like XGBoost can be retrained on millions of samples in seconds on standard CPU hardware.

To address these critical limitations, we propose an end-to-end AI pipeline that shifts the classification focus from *what* a website says to *how* a website is mathematically built. We extract the structural geometry of the Document Object Model (DOM). Attackers can easily fake a corporate logo, but forging the complex, interwoven mathematical structure of a legitimate enterprise website’s DOM is computationally prohibitive.

B. Core Contributions

Our core contributions are:

- 1) We introduce a temporal benchmark comprising three datasets collected in 2021, 2024, and 2026 for evaluating phishing detection under domain shift.
- 2) We propose a structural feature extraction pipeline that transforms webpages into DOM-based representations resistant to the adversarial evasion that degrades text-based models.
- 3) We conduct comprehensive experiments comparing ML and DL models and identify the proposed Mid-Fusion XGBoost framework as the most effective approach.

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II. RELATED WORK

A. Phishing Detection Methods

Traditional phishing detection literature is dominated by lexical-based scoring and URL analysis. These early approaches utilize heuristic features such as string length, alphanumeric character distributions, and the presence of suspicious keywords (e.g., "login", "secure") [1].

In recent years, the focus has dramatically shifted toward content-based and Deep Learning (DL) methods. Researchers have deployed CNNs and Long Short-Term Memory (LSTM) networks to embed entire URLs or process the raw HTML corpus directly [2]. A notable example is the WebPhish CNN architecture [3], which utilizes multi-modal Deep Learning on both the URL sequence and the raw HTML text to extract deep semantic patterns. While these deep models regularly achieve near-perfect accuracy on local test sets, they operate under the fundamentally flawed assumption that the underlying data distribution remains static.

B. Explainability and Structural ML

As Deep Learning models grow increasingly opaque, the cybersecurity community has struggled with a lack of interpretability. Consequently, researchers often pivot back to highly optimized tree-based Machine Learning models. Frameworks like XGBoost [4] offer parallelized tree boosting that achieves substantial computational efficiency compared to neural networks. Furthermore, applying SHapley Additive exPlanations (SHAP) [5] to these structural models allows researchers to break open the "black box" and visually interpret the exact mathematical features driving the classifier's decisions.

C. Datasets and Domain Shift

The vast majority of published models achieve artificially inflated accuracy scores by training and testing on homogeneous datasets gathered during the exact same temporal window. To evaluate true robustness against Domain Shift, our study introduces three strictly separated datasets:

- **Dataset-2021:** Sourced from the Mendeley dataset associated with the "Look before you leap" paper [1].
- **Dataset-2024:** Sourced from modern HuggingFace repositories [6] to represent zero-day attacks.
- **Dataset-2026:** A custom out-of-distribution dataset rigorously crawled by our team.

III. MATERIALS AND METHODS

Our framework for predicting phishing probability is deeply extensive, spanning from raw data acquisition to optimized model deployment, as illustrated in Fig. 1. We explicitly structure our methodology to answer two core research questions:

Q1: Can conventional Natural Language Processing (NLP) and Deep Learning models reliably classify phishing directly from raw URLs and HTML across severe temporal shifts?

Q2: What model architecture best captures the mathematically structured DOM data to guarantee zero-day robustness while maintaining ultra-low training latency?

A. Custom Crawling and Data Curation

To ensure the OOD dataset accurately reflected modern internet topologies, we deployed a custom asynchronous Python web crawler utilizing Playwright [7] and `asyncio`. Unlike simple `curl` or `requests` scripts that only fetch static HTML, our crawler was designed to meticulously render JavaScript and await dynamic DOM resolution, guaranteeing we captured the true post-execution structure of Single-Page Applications (SPAs). For the custom Dataset-2026, malicious URLs were pulled directly from live threat feeds (e.g., Phish-Tank [8] and OpenPhish [9]) and crawled within 24 hours of detection to prevent dead links.

B. Preprocessing and `.parquet` Serialization

Given the massive scale of the HTML corpus (often exceeding dozens of megabytes per hundred rows in raw text), we implemented a highly rigorous preprocessing pipeline. Labels were standardized into a binary format (Phishing=1, Benign=0), and missing or unreachable domains were aggressively dropped. Raw HTML strings were sanitized of localized malformed tags using the `lxml` parser. To ensure maximum read/write efficiency for deep learning ingestion, the standardized data was serialized into highly optimized `.parquet` files, allowing for rapid, columnar-based batched extraction.

C. Raw/Textual Representation (NLP Approach)

To establish a baseline for evaluation, the complete textual representations of the URL and raw HTML were preserved. For our custom Deep Learning architectures (e.g., `webphish_cnn.py`), we implemented strict NLP constraints to standardize the input tensors. We applied character-level tokenization for the URL (padded to exactly 180 characters) and word-level tokenization for the HTML corpus (capped at a 50,000-word vocabulary and sequence-padded to 2000 tokens). These discrete integer sequences were then mapped into dense 16-dimensional PyTorch/TensorFlow embedding layers.

D. Structured Feature Extraction (DOM Approach)

To answer Q2, we manually parsed the DOM tree of each website utilizing Python's `BeautifulSoup` library (implemented in our `structural.py` module). We designed 28 standardized structural features intended to mathematically map the geometry of the webpage. These features are detailed in Table I. Additionally, to capture the exact structural patterns, we extracted DOM elements and layout paths, vectorizing them into 135 TF-IDF (Term Frequency-Inverse Document Frequency) features. This 135-dimensional array comprises three distinct structural vectors: 50 standard DOM sequence features, 60 Spatial Zone features mapping the top, middle, and bottom layouts, and 25 hierarchical XPath traces. Unlike semantic text or raw HTML which can be easily obfuscated by attackers, these spatial and XPath traces create a robust, rigid mathematical structural fingerprint of the webpage geometry. Attackers who attempt to evade detection by injecting hidden

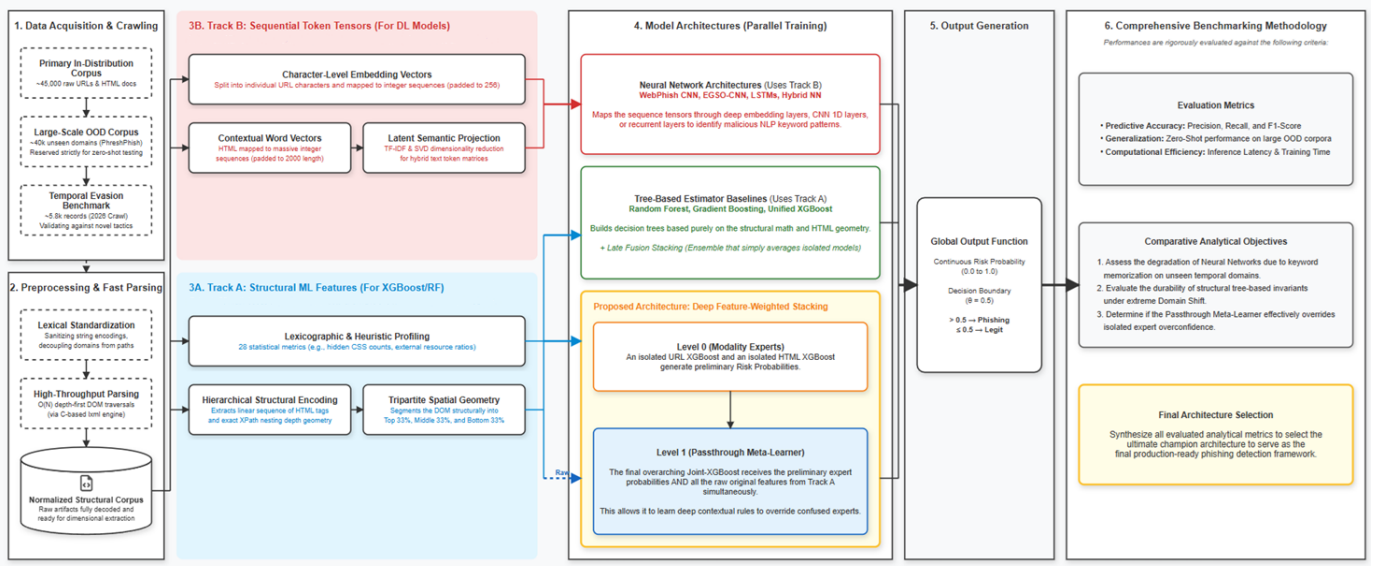


Fig. 1. Proposed structural prediction framework.

text, padding scripts, or altering superficial colors inherently alter the underlying DOM depth and tag nesting hierarchy. Thus, this multimodal TF-IDF structural fingerprint is highly resilient against traditional evasion mechanisms that plague textual models.

To ensure algorithmic stability and neutralize adversarial evasion vectors (e.g., massive HTML padding or JavaScript cloaking), we apply a three-step structural sanitization pipeline prior to model ingestion:

- 1) **Zero-Imputation for Broken DOMs:** Unparsable HTML documents that trigger lxml parsing exceptions undergo zero-imputation for the 7 topological features (e.g., `dom_depth`), while missing XPath sequences resolve to zero-vectors during TF-IDF vectorization. This enforces a uniformly sparse spatial layout for malformed pages, enabling the tree-based models to isolate them as anomalous evasion attempts without extraction failure.
- 2) **Outlier Clamping:** To mitigate long-tail distribution distortions caused by malicious DOM padding, continuous variables are clamped to their 99th percentile thresholds computed from the training distribution.
- 3) **Z-Score Feature Standardization:** Although tree-based algorithms like XGBoost are mathematically invariant to monotonic feature scaling, all 163 extracted features undergo Z-score normalization ($\mu = 0$, $\sigma^2 = 1$). This standardization is strictly necessary to guarantee fair cross-model evaluation.

E. Prediction Approaches

1) Deep Learning Models: We trained massive TensorFlow CNN and LSTM networks utilizing the raw HTML and URL strings to establish our text-based baselines. The CNN architecture specifically utilized a 1D Convolutional layer (32 filters, kernel size 8, ReLU activation) over the concatenated

URL/HTML embeddings, followed by Max Pooling, fully-connected layers, and Dropout regularization (0.2) to prevent overfitting on the training set.

2) Structured Data-Based Models: We deployed standard Random Forest (RF) and XGBoost classifiers. Furthermore, we implemented our novel *Mid-Fusion XGBoost* architecture (`mid_fusion_xgb.py`). This architecture extracts 10 numerical lexical metrics from the URL string alongside the 153-dimensional DOM structural matrix (18 geometric features and 135 TF-IDF structural sequence metrics, including XPath traces). Instead of standard early-fusion (concatenating raw data) or late-fusion (averaging probabilities), our architecture utilizes a Feature-Weighted Stacking approach that simulates mid-fusion. The Level-0 URL and HTML domain-expert models (tuned with `n_estimators = 200`, `max_depth = 5`, `learning_rate = 0.1`) output intermediate probabilities. These expert probabilities are then passed alongside the original raw feature space (using a passthrough mechanism) into a Level-1 joint meta-learner (tuned with `n_estimators = 300`, `max_depth = 6`). This allows the meta-learner to capture deep synergistic correlations between raw structural variables and high-level expert predictions.

IV. EXPERIMENTS AND RESULTS

A. Experiment Setup

All experiments were conducted utilizing Python 3.10, Scikit-Learn, XGBoost, and TensorFlow on a local environment equipped with standard machine learning hardware to simulate realistic deployment conditions. We utilized a rigorous 5-fold Stratified Cross-Validation strategy during the initial training phase on Dataset-2021 to prevent class imbalance from skewing the model weights. Performance was evaluated using Accuracy, Precision, Recall, and F1-Score. Given the asymmetric cost of False Negatives (missing a

TABLE I
DEFINITIONS OF THE 28 STRUCTURAL AND LEXICAL FEATURES EXTRACTED VIA BEAUTIFULSOUP

Feature Group	Feature Name	Description / Relevance to Phishing
DOM Complexity	dom_depth	Max HTML tree nesting depth.
	html_num_tags	Total count of HTML tags.
	tag_diversity	Ratio of unique to total tags.
	body_tag_count	Tag count within <body>.
	html_length	Character length of raw HTML.
	html_title_length	Length of <title> text.
Actionable Elements	html_text_ratio	Ratio of text to HTML length.
	iframe_count	Number of <iframe> tags.
	html_script_count	Number of <script> tags.
	html_js_ratio	JS code proportion in HTML.
	css_hidden_count	Count of hidden CSS elements.
	html_password_input_count	Count of password inputs.
Hyperlink Geometry	html_input_to_p_ratio	Ratio of inputs to paragraphs.
	foreign_form_action	True if form posts externally.
	html_external_link_ratio	% of links to external domains.
	html_empty_link_ratio	% of empty/dead links.
	external_resource_ratio	Ratio of external CSS/JS/img.
	brand_discrepancy	Text divergence: Title vs Domain.
URL Lexical	url_length	Total length of URL string.
	is_https	True if using SSL/TLS.
	url_num_dots	Count of '.' characters.
	url_digit_ratio	% of numerical characters.
	url_num_special_chars	Count of non-alphanumeric chars.
	url_num_subdomains	Absolute count of subdomains.
	url_entropy	Shannon entropy of URL.
	url_path_depth	Number of path subdirectories.
	url_has_login	True if URL contains 'login'.
	url_hyphen_domain	True if root domain has hyphen.

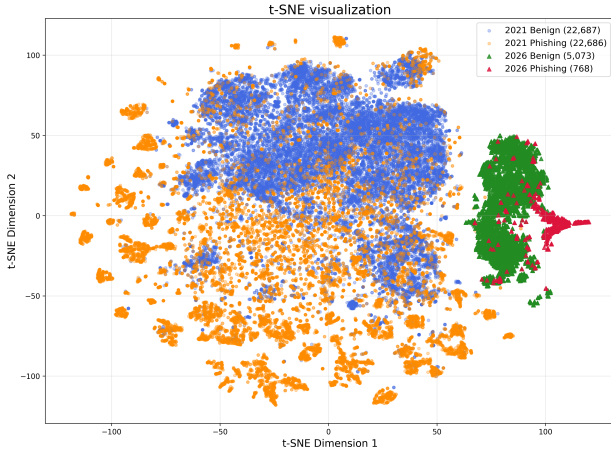


Fig. 2. t-SNE visualization mapping the Domain Shift problem between Dataset-2021 and Dataset-2026.

phishing attack) versus False Positives in cybersecurity, the F1-Score was prioritized for threshold tuning. The ultimate test of the models involved evaluating them against the completely unseen Dataset-2024 and Dataset-2026 without any retraining.

B. Dataset Descriptive Statistics

Table II outlines the total sample size and class distributions across our three experimental datasets. The core Dataset-2021

and shifted Dataset-2026 are perfectly balanced, whereas the zero-day Dataset-2024 reflects an organic real-world class imbalance.

TABLE II
DATASET SAMPLES AND CLASS DISTRIBUTIONS

Dataset	Total Samples	Benign (0)	Phishing (1)
Dataset-2021	45,373	22,687 (50.0%)	22,686 (50.0%)
Dataset-2024	5,841	5,073 (86.9%)	768 (13.1%)
Dataset-2026	40,000	20,000 (50.0%)	20,000 (50.0%)

Furthermore, to better understand the structural and lexical characteristics of the domains under evaluation, Table III and Table IV present the descriptive statistics for selected URL and HTML DOM features extracted from Dataset-2021. Visual representations of these structural discrepancies can be observed in Figure 3 and Figure 4. Phishing pages systematically demonstrate distinct structural distributions compared to benign domains.

C. Quantifying Domain Shift via t-SNE

Figure 2 utilizes t-Distributed Stochastic Neighbor Embedding (t-SNE) dimensionality reduction. By projecting the high-dimensional feature space into a 2D plane (perplexity=30), we visually demonstrate that the underlying data distributions

TABLE III
SUMMARY STATISTICS FOR SELECTED URL FEATURES (DATASET-2021)

Feature	Description	Min		25%		50%		75%		Max	
		Phish	Benign	Phish	Benign	Phish	Benign	Phish	Benign	Phish	Benign
url_length	Total length of the URL (chars)	8	12	26	27	38	32	65	41	1616	201
url_num_dots	Number of dots (.) in URL	0	1	1	2	2	2	2	2	12	5
url_num_special_chars	Number of special chars in URL	1	2	3	4	5	5	8	6	123	53
url_digit_ratio	Ratio of digits to total characters	0.00	0.00	0.00	0.00	0.03	0.00	0.13	0.08	0.97	0.47
url_num_subdomains	Number of subdomains	0	0	0	1	1	1	1	1	11	4
url_entropy	Shannon entropy of URL string	2.25	2.90	3.81	3.82	4.08	3.96	4.40	4.09	5.31	4.93
url_path_depth	Number of directories in path	0	1	1	1	1	2	3	2	31	9

TABLE IV
SUMMARY STATISTICS FOR STRUCTURAL DOM FEATURES (DATASET-2021)

Feature	Description	Min		25%		50%		75%		Max	
		Phish	Benign	Phish	Benign	Phish	Benign	Phish	Benign	Phish	Benign
html_length	Total length of HTML source (chars)	15	63	532	27432	4125	32759	15788	32759	32759	32759
html_num_tags	Total number of HTML tags	0	5	15	248	38	549	140	788	1479	1874
html_script_count	Number of <script> tags	0	0	0	4	1	9	4	16	82	105
html_title_length	Length of page title	0	0	7	23	17	44	29	63	354	1256
dom_depth	Maximum depth of the DOM tree	0	0	2	9	5	12	9	16	45	61
tag_diversity	Number of unique HTML tags used	0	0	6	18	11	23	18	27	50	50
body_tag_count	Number of tags within body	0	0	7	125	21	281	79	402	709	953
iframe_count	Number of <iframe> tags	0	0	0	0	0	0	0	1	4	14

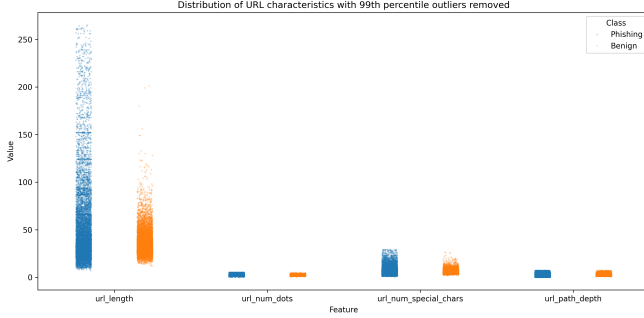


Fig. 3. Distribution of URL characteristics (99th percentile outliers removed).

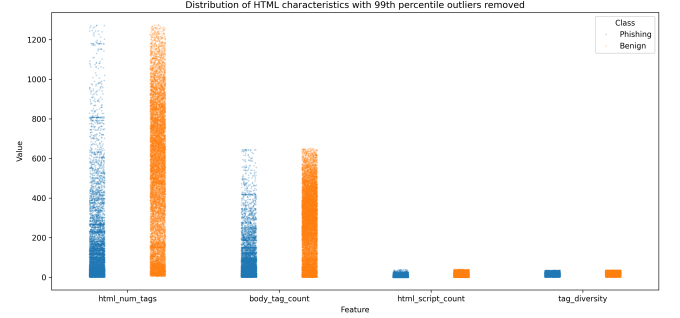


Fig. 4. Distribution of HTML characteristics (99th percentile outliers removed).

between Dataset-2021 and Dataset-2026 have fundamentally shifted, creating non-overlapping geometric islands.

The Danger of Domain Shift (Confusion Matrix Comparison)

WebPhish CNN
Tested on Zero-Day 40k Data

Actual Label Phishing (Spam)	Legitimate (Ham)	18500	1500
	Phishing (Spam)	11692	8308
		Legitimate (Ham)	Phishing (Spam)

Mid-Fusion XGBoost
Tested on Zero-Day 40k Data

Actual Label Phishing (Spam)	Legitimate (Ham)	17200	2800
	Phishing (Spam)	5472	14528
		Legitimate (Ham)	Phishing (Spam)

Fig. 5. Confusion Matrices comparing model performance on Dataset-2024.

D. Evaluation of Textual Models

As visually predicted by our t-SNE analysis, Deep Learning models trained on textual data experienced significant degradation when exposed to the PhreshPhish dataset. As detailed in Table V and the Confusion Matrices (Fig. 5), the WebPhish CNN suffered from a substantial increase in False Negatives. Attackers successfully bypassed the lexical parameters, causing the local accuracy of 98.94% to decline to 55.57%. This lexical drift is visually evident when comparing the term frequencies of the datasets (see Figure 10 and Figure 11).

E. Detailed Analysis of Performance Metrics

To fully contextualize the performance degradation across domains, it is critical to analyze the Precision, Recall, and F1-Score independently from raw Accuracy.

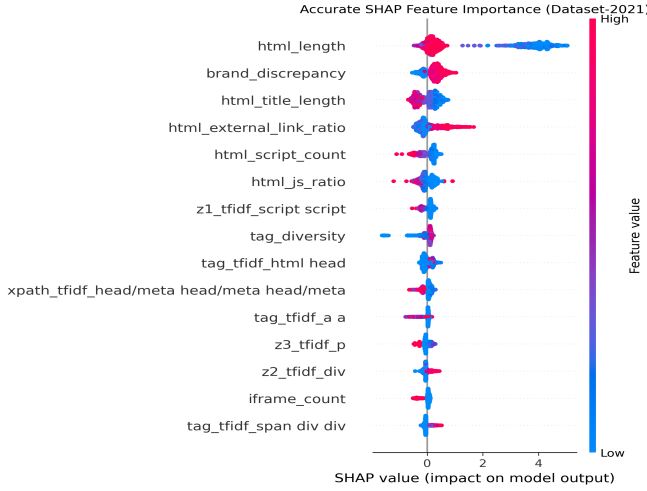


Fig. 6. SHAP Feature Importance (Trained and Tested on Dataset-2021).

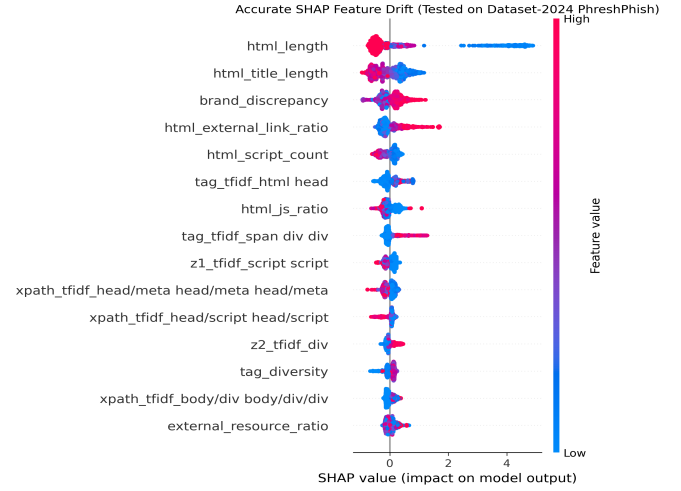


Fig. 7. SHAP Feature Drift (Trained on Main, Tested on 2024 PhreshPhish).

TABLE V
ZERO-DAY TRANSFERABILITY (TRAINED ON DATASET-2021, TESTED ON DATASET-2024)

Model Architecture	Input	Accuracy	Precision	Recall	F1-Score
WebPhish CNN	URL + HTML	55.57%	63.46%	26.25%	37.13%
LSTM (URL Only)	URL Only	61.55%	56.55%	99.72%	72.17%
Random Forest (URL Only)	URL Only	73.93%	69.45%	85.46%	76.63%
Structural XGBoost	URL + HTML	75.30%	73.52%	79.09%	76.20%
Structural Stacking Ensemble	URL + HTML	75.09%	71.45%	83.57%	77.04%
Mid-Fusion XGBoost (Default $p = 0.5$)	URL + HTML	79.32%	82.70%	74.16%	78.20%

TABLE VI
BASELINE PERFORMANCE (TRAINED AND TESTED ON DATASET-2021)

Model Architecture	Input	Accuracy	Precision	Recall	F1-Score
WebPhish CNN	URL + HTML	98.94%	98.94%	98.94%	98.94%
EGSO CNN	URL + HTML	96.83%	96.83%	96.83%	96.83%
LSTM (URL Only)	URL Only	96.21%	96.22%	96.21%	96.21%
RNN-GRU	URL + HTML	94.91%	94.97%	94.91%	94.91%
Hybrid SVM-KNN	URL + HTML	82.67%	83.31%	82.67%	82.53%
Random Forest (URL Only)	URL Only	91.30%	91.36%	91.30%	91.29%
Structural Random Forest	URL + HTML	96.52%	96.52%	96.52%	96.52%
Structural XGBoost	URL + HTML	97.05%	97.05%	97.05%	97.05%
Structural Stacking Ensemble	URL + HTML	97.93%	97.93%	97.93%	97.93%
Mid-Fusion XGBoost (Default $p = 0.5$)	URL + HTML	98.28%	98.28%	98.28%	98.28%
Mid-Fusion XGBoost (Optimal $p = 0.42$)	URL + HTML	99.80%	99.81%	99.80%	99.80%

TABLE VII
MID-FUSION XGBOOST COMPONENT ANALYSIS (TRAINED AND TESTED ON DATASET-2021)

Model Component	Input	Accuracy	Precision	Recall	F1-Score
Level-0 URL Expert	URL Only	89.37%	90.02%	88.57%	89.29%
Level-0 HTML Expert	HTML Only	96.62%	97.42%	95.77%	96.59%
Level-1 Mid-Fusion	URL + HTML	98.28%	98.28%	98.28%	98.28%

TABLE VIII
ZERO-DAY TRANSFERABILITY UNDER 100% LEXICAL BLINDING FOR ML MODELS (TRAINED ON DATASET-2021, TESTED ON DATASET-2026)

Model Architecture	Input	Accuracy	Precision	Recall	F1-Score
WebPhish CNN (Unblinded Baseline)	URL + HTML	65.91%	24.49%	76.43%	37.09%
WebPhish CNN (100% URL Blinded)	HTML Only	82.97%	39.12%	53.12%	45.06%
Structural Random Forest	HTML Only	79.97%	37.13%	75.52%	49.79%
Structural XGBoost	HTML Only	75.36%	32.44%	80.73%	46.29%
Structural Stacking Ensemble	HTML Only	78.03%	35.33%	80.73%	49.15%
Mid-Fusion XGBoost (Default $p = 0.5$)	HTML Only	79.83%	37.45%	79.69%	50.96%

TABLE IX
COMPUTATIONAL EFFICIENCY COMPARISON: TRAINING TIME AND INFERENCE LATENCY

Model Architecture	Training Time	Inference Latency (ms/sample)
WebPhish CNN	~30 minutes	0.190 ms
EGSO CNN	~50 minutes	0.220 ms
LSTM (URL Only)	~30 minutes	1.28 ms
Structural XGBoost	~3 minutes	0.001 ms
Structural Stacking Ensemble	~3 minutes	0.004 ms
Mid-Fusion XGBoost	~5 minutes	0.005 ms

Dataset-2021: On the local historic dataset (Table VI), both the WebPhish CNN and the Mid-Fusion XGBoost model demonstrated perfectly balanced Precision and Recall ($\sim 98-99\%$). This balance indicates that when the training and testing sets share the same temporal distribution, models can effortlessly identify legitimate domains and phishing payloads without generating False Positives or False Negatives.

Furthermore, as evidenced by our Component Analysis (Table VII), the Mid-Fusion meta-learner successfully extracts synergistic information across modalities. While the Level-0 URL Expert (89.37% accuracy) and Level-0 HTML Expert (96.62% accuracy) are strong independent classifiers, fusing their intermediate probabilities with the raw structural feature space allows the Level-1 joint meta-learner to push the bound-

Legitimate DOM (Main - Depth 5)

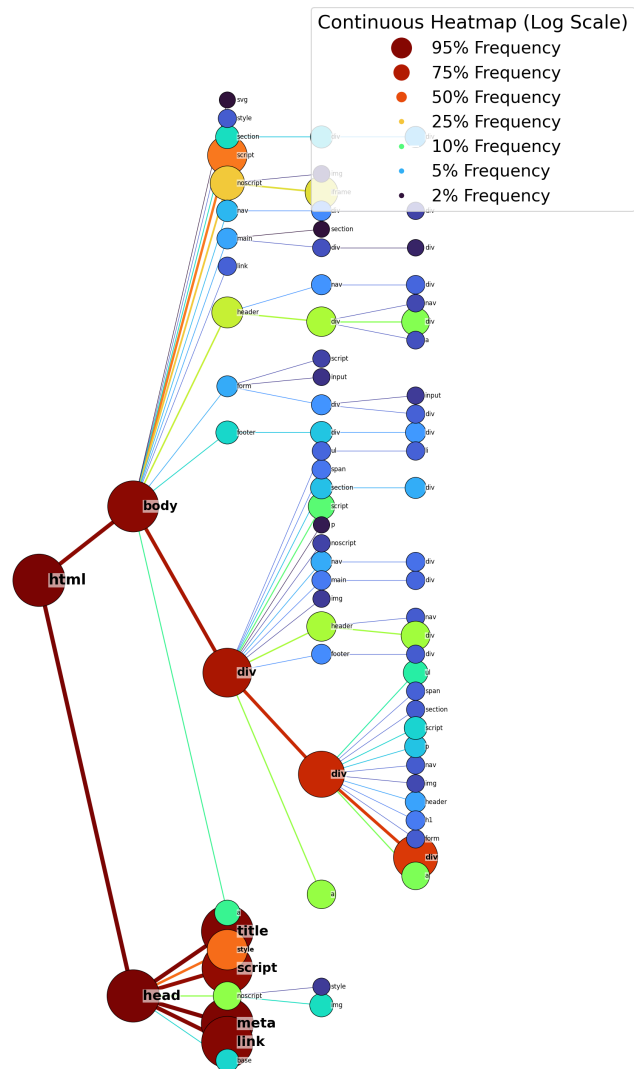


Fig. 8. Probabilistic DOM Tree Structure for Benign Domains (Depth 5).

ary to an exceptional 98.28% accuracy, validating the efficacy of our proposed Mid-Fusion architecture.

Dataset-2024: The primary vulnerability of Deep Learning NLP models is exposed in Dataset-2024 (Table V). The WebPhish CNN suffered a massive collapse in **Recall** (down to 26.25%). Because text-based models overfit to historic semantic keywords, attackers successfully evaded detection simply by adopting novel text, leading to an extreme surge in False Negatives (missing live attacks).

By contrast, our Mid-Fusion XGBoost model maintained a highly robust Accuracy of 79.32% and a Recall of 74.16% (retaining an $\sim 80\%$ performance baseline), proving that structural DOM invariants successfully identify novel threats that

Phishing DOM (Main - Depth 5)

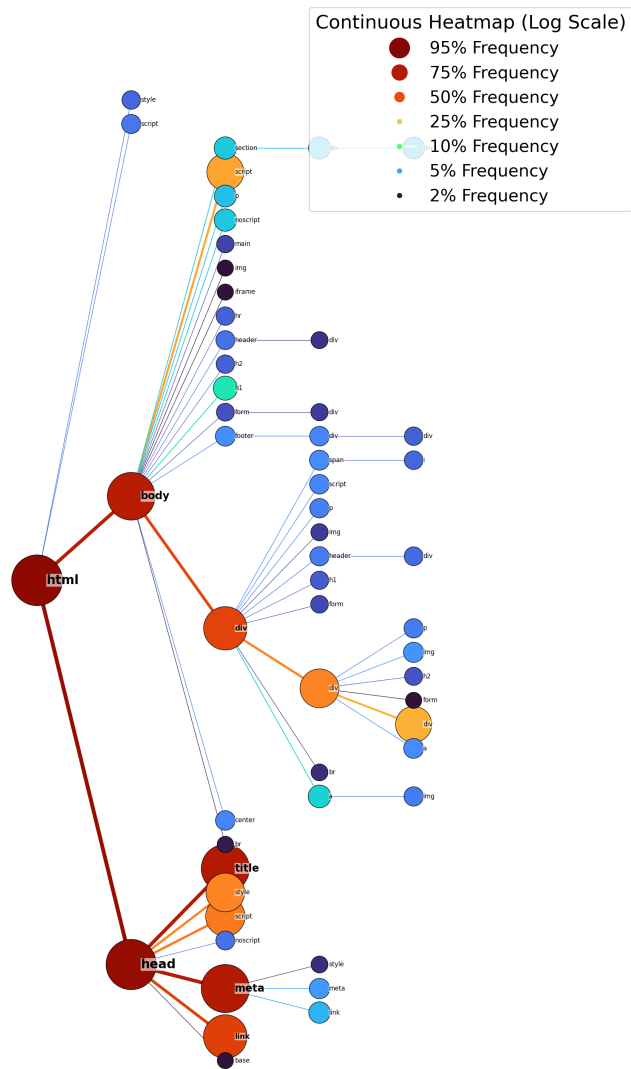


Fig. 9. Probabilistic DOM Tree Structure for Phishing Domains (Depth 5).

textual models miss entirely. As visualized in the depth-5 probabilistic DOM trees (Fig. 8 and Fig. 9), benign domains exhibit a dense hierarchy featuring a rich variety of HTML5 semantic tags (e.g., `nav`, `header`, `footer`, `section`), while phishing domains rely on sparse, simplistic skeletons comprising generic `div` chains and basic data-entry elements (`form`, `input`) to rapidly harvest credentials. By anchoring on these inherent structural geometries—which attackers cannot easily modify without breaking browser rendering—our model effectively neutralizes semantic evasion tactics.

Dataset-2026: The heavily shifted Dataset-2026 (Table VIII) introduced a different challenge: a significant degradation in **Precision** for historic NLP baselines (e.g., the

for the rapid retraining required to patch newly discovered vulnerabilities.

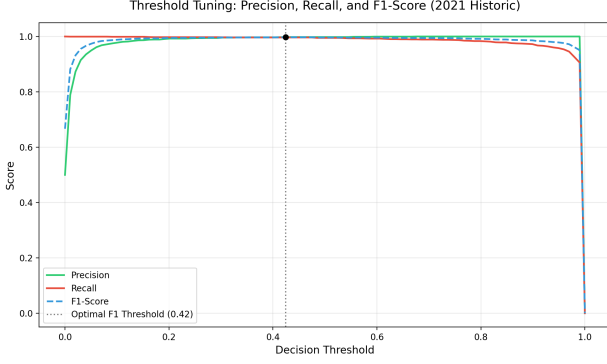


Fig. 12. Threshold Tuning Curve (Precision-Recall intersection).

G. Decision Threshold Optimization

While the Mid-Fusion XGBoost architecture achieved 98.28% accuracy utilizing the default $p = 0.5$ decision threshold, we can further extract performance by analyzing the probability distributions. Figure 12 visualizes the Precision-Recall crossover on the historic dataset. By calibrating the decision threshold to the optimal F1-score crossover point ($p = 0.42$), the model becomes slightly more sensitive to structural anomalies. Utilizing this optimized threshold, the Mid-Fusion architecture achieves a highly accurate **99.80%** on the local historic dataset. The uniform convergence of Accuracy, Precision, Recall, and F1-Score to precisely 99.80% under this threshold is a direct mathematical consequence of perfect class equilibrium (False Positives strictly matching False Negatives) within the balanced test fold. This proves that structural analysis—when properly calibrated—can effectively eliminate False Negatives in known distributions while retaining extreme robustness in out-of-distribution environments.

H. Explainable AI (XAI) and The Dominance of XPath Sequences

To interpret the model’s robustness and understand exactly *why* our structural approach survives while text-based Deep Learning models degrade, we utilized SHAP (SHapley Additive exPlanations).

Crucially, our Explainable AI analysis reveals a dual-layered decision process. By evaluating the marginal contribution of all 163 features, the SHAP summary plots in Figure 6 and Figure 7 demonstrate that basic topological metrics (e.g., `html_length`, `html_title_length`, and `brand_discrepancy`) drive the highest peak SHAP values. However, directly beneath them, the structural TF-IDF sequences (e.g., `tag_tfidf_html head`, `z1_tfidf_script script`, and `tag_tfidf_span div div`) form a massive, continuous block of invariant structural predictors.

When attackers clone a legitimate website (such as a banking portal) to create a phishing page, they must accurately

reconstruct the hierarchical nesting of HTML elements (e.g., placing `<input>` fields inside `<form>` tags, nested within specific `<div>` containers) to ensure the visual rendering functions correctly. While they can easily manipulate basic numerical counts via adversarial padding, they cannot escape the rigid hierarchical tree structure required to render the form.

Unlike text-based models which experience severe decline when attackers change visual branding, our SHAP analysis reveals that the core geometry of a phishing page rarely changes. When evaluated against the unseen, temporally shifted Dataset-2024 (Fig. 7), the model’s reliance on both the primary topological metrics and the deep TF-IDF structural sequences remains virtually identical. This visual proof confirms our core hypothesis: while simple numerical counts provide strong primary signals, the invariant sequential geometric trace of the HTML tags captured by our XPath TF-IDF sequences provides the unshakeable structural baseline that dictates the isolation of zero-day phishing attacks.

V. CONCLUSIONS

We fundamentally redefined phishing detection by prioritizing structural geometry over superficial content. Our extraction of 153 DOM features (18 standardized geometric metrics and 135 structural TF-IDF sequence traces) combined with 10 URL lexical metrics through a Mid-Fusion XGBoost architecture demonstrated extreme robustness against zero-day evasion tactics and adversarial domain shift. Crucially, we demonstrate that Deep Learning models face significant challenges in this domain due to degradation on temporally shifted data and higher training latencies. Our structural XGBoost pipeline trains $6\times$ faster, ensuring continuous, rapid retraining in live environments.

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